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Timing of Rotational Motion of Southwest Japan Inferred from Paleomagnetism of the Setouchi Miocene Series

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Paleomagnetic measurements have been made on tuff and mud samples from 25 sites of the Morozaki Group and the Mizunami Group, distributed in the eastern part of Southwest Japan. Stability tests by alternating field and thermal demagnetization reveal the existence of characteristic remanent magnetization, of which directions show significant clockwise deflection; the mean declination shift exceeds 50° in the late Early Miocene members (N6–N8), and is less than 30° in the early Middle Miocene members (N9). This result, compared with the other paleomagnetic data from the Setouchi Miocene Series and the volcanic rocks in the San'in district, suggests that coherent rotation of Southwest Japan began between the late Early Miocene and the early Middle Miocene. Nearly a half of the rotation might have been attained in the stage of Zone N9, which possibly corresponds to the period of the opening and increasing subsidence in the Japan Sea.

1. Introduction

Recent paleomagnetic studies (OTOFUJI and MATSUDA, 1983; 1984; HAYASHIDA and ITO, 1984; TORII, 1985) have revealed that Southwest Japan suffered a clockwise rotation at about 15 Ma. The timing of the rotation is much later than previously assumed (KAWAI *et al.*, 1971), and the duration of the rotation seems extremely short. Current research is directed toward increasing the number of paleomagnetic data near the timing of the rotation. This is important for more detailed description of the rotational process, which has a significance in reconstruction of tectonic history around Southwest Japan.

Paleomagnetic measurements have been made on samples from the Morozaki Group and the Mizunami Group, typical sequences of the Setouchi Miocene Series sporadically extending in Southwest Japan (Fig. 1). Among them, the late Early Miocene Ichishi Group was shown to have remanent magnetization having easterly deflected declinations (HAYASHIDA and ITO, 1984). The age of the Ichishi Group, about 16 Ma, gives a critical timing for the beginning of the rotation of Southwest Japan. While the Morozaki and Mizunami Groups are correlative to the Ichishi Group in part, the upper members of both groups are suggested to be younger than the Ichishi Group. The paleomagnetic data from these sections are, therefore, suitable for the purpose mentioned above. Additionally, comparison of the paleomagnetic directions between the sedimentary basins of the Setouchi Miocene Series would contribute to assessment for dimension of the rotated block.

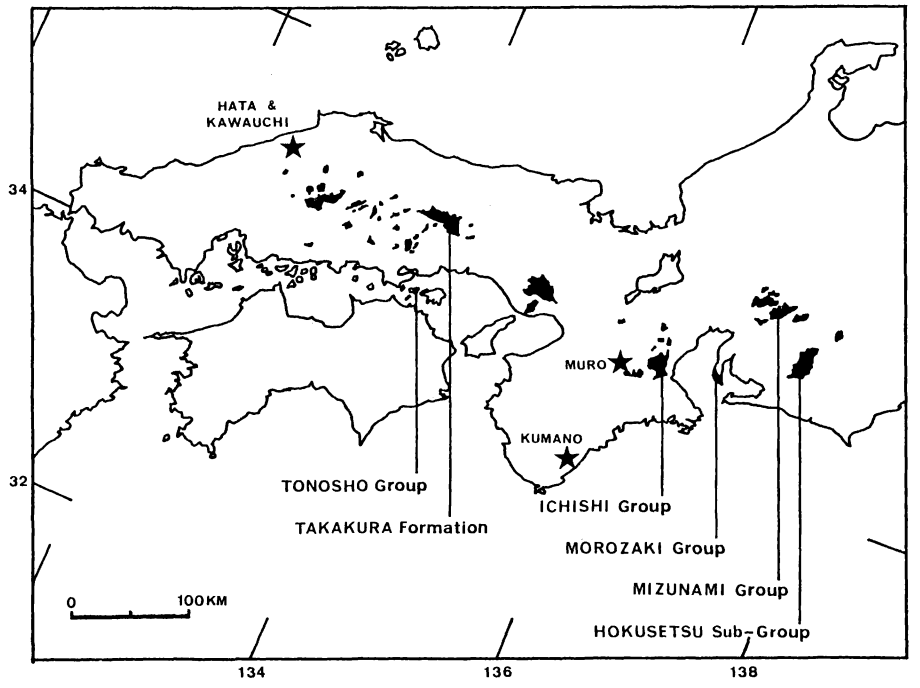


Fig. 1. Map showing the distribution of the Setouchi Miocene Series. The Morozaki Group and the Mizunami Group are located in the eastern part of Southwest Japan. Location of some volcanic rocks referred in the text are shown with stars.

2. Background

The past paleomagnetic studies done before 1970's have already revealed that remanent magnetization of the Cretaceous and Paleogene rocks in Southwest Japan shows significant declination shift relative to those in Northeast Japan (KAWAI *et al.*, 1961; 1971), and to those in the Korean Peninsula (YASKAWA, 1975). The pioneer workers suggested that the declination shift was caused by a clockwise rotation of Southwest Japan. The rotation was assumed to have occurred in the Paleogene time, possibly associated with opening of the Japan Sea (e.g., UYEDA and MIYASHIRO, 1974).

Recent researches (OTOFUJI and MATSUDA, 1983; 1984; HAYASHIDA and ITO, 1984; TORII, 1985) have elaborated to determine timing of the rotational motion. OTOFUJI and MATSUDA (1983) made a paleomagnetic survey of the Cretaceous to Miocene igneous rocks distributed in the west of the San'in district, and showed that the easterly deflected remanent magnetizations are observed in the latest Oligocene rocks dated at 28 Ma as well as in the older members. It was thus suggested that the rotation of Southwest Japan had not occurred during the Paleogene time and that the Japan Sea was opened possibly in the Miocene. HAYASHIDA and ITO (1984) revealed that the Ichishi Group, dated at about 16 Ma, also shows clockwise declination shift of remanent magnetization. Based on the paleomagnetic direction of the late Early

Miocene Ichishi Group and the other paleomagnetic data from the radiometrically dated igneous rocks (HIROOKA *et al.*, 1972; TORII, 1979; TAGAMI, 1982; TOSHA and TSUNAKAWA, 1981), HAYASHIDA and ITO (1984) suggested that Southwest Japan had been situated adjacent to the Korean Peninsula until 16 Ma, and then suffered a clockwise rotation. The same idea was drawn by OTOFUJI and MATSUDA (1984), who showed that the volcanic rocks in the San'in district, dated at about 20 Ma, have the remanent directions deflected to the east. The 'undeflected' paleomagnetic directions, which suggest the accomplishment of the rotation, were reported by TORII (1985) from the Setouchi volcanic rocks. The Setouchi volcanic rocks overlie the Setouchi Miocene Series in some areas, and their radiometric dates range from about 14 Ma to 11 Ma. The age of the Setouchi volcanics gives timing for the termination of the rotation of Southwest Japan.

OTOFUJI *et al.* (1985) compiled a reliable paleomagnetic data-set from Southwest Japan for the period between 40 Ma and the present. They proposed a hypothetical model that the clockwise rotation through about 47° occurred during the period about 1 m.y. at 14.9 Ma. The rotated block was assumed to be of a micro-plate size, extending from the San'in area in the west and to the Shidara basin in the east. Suggesting that the back-arc spreading is the most probable cause of the rotation of Southwest Japan, OTOFUJI *et al.* (1985) regarded the timing of the rotation as that of the opening of the Japan Sea. In their model, the rapid rotation of the Southwest Japan arc requires very large rate of sea-floor spreading in the Japan Sea.

The timing of the rotation of Southwest Japan determined by OTOFUJI *et al.* (1985) is mainly constrained from radiometric dates of igneous rocks. The felsic igneous rocks, such as the Kumano Acidic Rocks and the Muro Volcanics, have deflected magnetic directions, and yield the K–Ar and fission track dates ranging around 16 to 15 Ma. On one hand, the Setouchi volcanic rocks, of which magnetizations are in the north-south direction, are dated as younger than about 14 Ma (TORII, 1985). These data suggest termination of the rotational motion through over 45° within just about 1 m.y. The ambiguity of the radiometric dates, however, seems to exceed the duration of the rotation, and this means that the process of the rapid rotation is not fully described through compilation of the paleomagnetic data from radiometrically dated rocks. Instead, paleomagnetic studies of continuous sequences of strata may provide more complete records of the rotation process. Correlating the surveyed sections to a standard timescale of magnetostratigraphy and biostratigraphy, such records would constitute other lines of evidence for the rapid rotation of Southwest Japan.

3. Geological Setting

Paleomagnetism of the Setouchi Miocene Series is of great significance in more detailed description of the rotational motion of Southwest Japan, because they are assigned to the late Early Miocene to the early Middle Miocene in age (ITOIGAWA, 1981). It has been shown that some sequences of the Setouchi Miocene Series, including the Morozaki and the Mizunami Groups, were deposited from the late Early Miocene to the early Middle Miocene (TSUCHI, 1981). The lower parts of the Morozaki and the Mizunami Groups are correlative to the late Early Miocene Ichishi Group. The upper members, assigned to the earliest Middle Miocene, are possibly

correspond to the stage between the Ichishi Group which has the deflected paleomagnetic directions (HAYASHIDA and ITO, 1984) and the Setouchi volcanics which show no significant rotation (TORII, 1985). The paleomagnetic data from such sections are, therefore, expected to give a more critical timing for the commencement of the rotation.

The Morozaki Group is distributed in the southern part of the Chita Peninsula, to the east of the Ichishi Group about 50 km apart (Fig. 1). Nonconformably overlying the basement Ryoke Complex, the Morozaki Group has a thickness about 1100 m. It is divided into three formations: Hikama, Toyohama, and Utsumi, in ascending order (DOI, 1983). The Hikama Formation is mainly composed of coarse sandstones, and alternations of sandstone and siltstone layers. The Toyohama Formation is rich in siltstones and is characterized by frequent intercalations of felsic volcanic ash layers. The Utsumi Formation, which was called the Minamichita Formation by IBARAKI *et al.* (1984), consists of alternations of tuffaceous sandstones and mudstones, also intercalating volcanic ash layers.

The geological age of the Morozaki Group was estimated using planktonic foraminiferal fossils. DOI (1983) reported occurrence of *Orbulina universa* in the upper part of the Utsumi Formation. It is acceptable that at least the topmost part of the Morozaki Group is assigned to the Middle Miocene. The older assemblages of planktonic foraminifers were found by IBARAKI *et al.* (1984) from two horizons of the Toyohama Formation. IBARAKI *et al.* (1984) suggested that these horizons are included in the N6 and N7 planktonic foraminiferal zones of BLOW (1969). Thus, the Morozaki Group is assigned to the Zones N6 to N9, corresponding to the late Early Miocene to the early Middle Miocene.

The Setouchi Miocene Series in the Mizunami area, about 80 km north of Morozaki, are distributed in three basins, Kani, Toki and Iwamuro basins, surrounded by basement rocks of the Mino Terrane and the Ryoke Complex (Fig. 1). The Mizunami Group in the Toki basin has been studied by many stratigraphers and paleontologists, and is one of the standard sections of the Setouchi Miocene. Comparing with the Morozaki Group, the Mizunami Group has a rather small thickness, about 500 m, and shows various lithofacies. ITOIGAWA (1974) described lithostratigraphy of the Mizunami Group in detail, dividing it into four formations: Toki lignite-bearing Formation, Hongo Formation, Akeyo Formation and Oidawara Formation, from lower to upper. The Toki lignite-bearing Formation is composed of mudstones, sandstones and conglomerates, which are supposed to have formed in a fresh-water environment. The Hongo Formation, which shows a remarkably tuffaceous lithology, and the Akeyo Formation were suggested to be products of a single cycle of marine transgression. The latter unit is disconformably overlain by the Oidawara Formation, which was assumed to have been formed during the following transgression.

ITOIGAWA and SHIBATA (1973) made analysis of sedimentary environment of the Setouchi Miocene Series based on molluscan assemblages, and suggested that the two phases of marine transgression in the Mizunami Group correspond to those observed in the upper part of the Morozaki Group. This correlation is supported by biostratigraphic data of micro-fossils. Planktonic foraminifera assemblages found in the Akeyo Formation suggest the correlation to the Zone N8 (SAITO, 1963; IBARAKI, 1981). The Oidawara Formation was assigned to the *Denticulopsis lauta* Zone of

diatom biostratigraphy (KOIZUMI, 1981), which is assumed to correspond to the Blow's Zone N9 (KOIZUMI, 1985; Koizumi, personal communication). Thus, the lower to middle part of the Mizunami Group is assigned to the late Early Miocene, and the Oidawara Formation in the upper part is to the early Middle Miocene.

4. Method and Results

Samples for paleomagnetic analysis were collected at 14 sites from the Morozaki Group, and at 11 sites from the Mizunami Group. Figure 2 shows stratigraphic

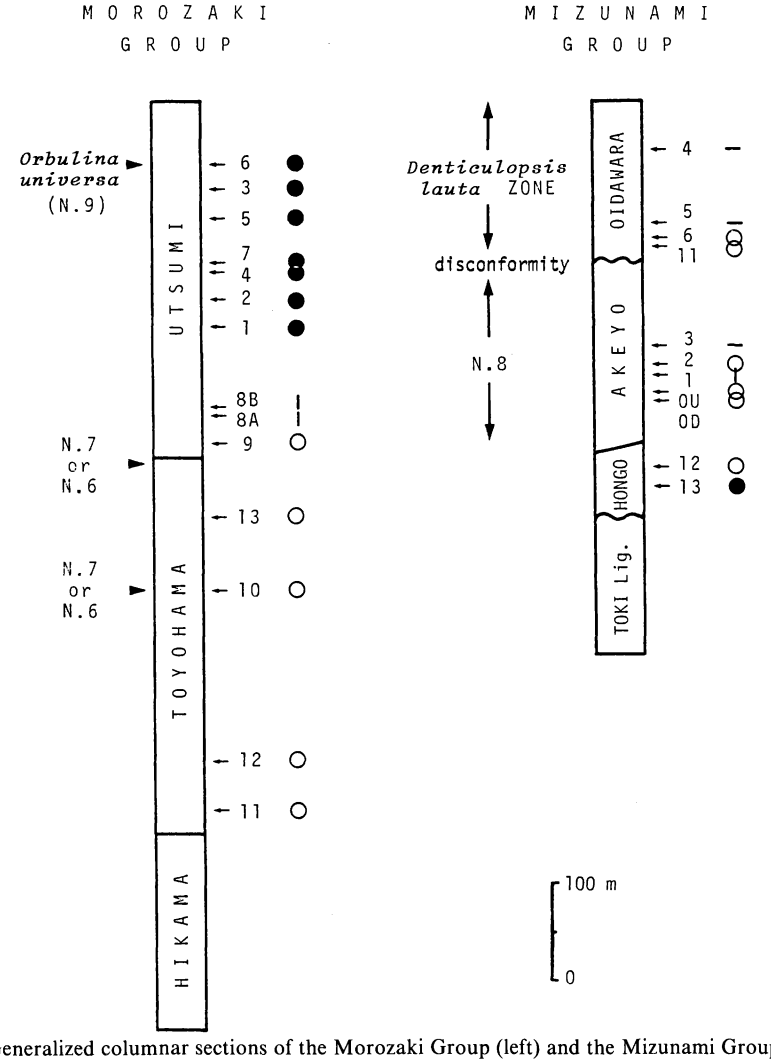


Fig. 2. Generalized columnar sections of the Morozaki Group (left) and the Mizunami Group (right). The columns are subdivided into formations, showing biostratigraphic data on the left side of them, and horizons of paleomagnetic sites. Magnetic polarities determined are also shown; solid circles: normal, open circles: reversed, I: intermediate, and—: undetermined.

positions of the sampling sites. The Hikama Formation of the Morozaki Group and the Toki lignite-bearing Formation of the Mizunami Group were not sampled, because coarse sandstones and conglomerates are dominant in these formations. At each sampling site, 6 to 10 oriented hand samples were obtained from a tuff or a mudstone layer. Strike and dip of the strata were measured for tilt correction of the remanent magnetic direction. Most of the sampling sites show gentle structural attitude, the strata dipping less than 15°.

Preliminary paleomagnetic data of 6 sites of the Utsumi Formation of the Morozaki were already reported by HAYASHIDA (1985). The preliminary study was carried out using the thermal demagnetization apparatus in which residual field over 100 nT remains. The residual field of this level seems to have prevented a determination of high-temperature magnetic component in some samples. The 6 sites of the Morozaki Group were again investigated in the present study.

In the laboratory, two or three cylindrical specimens were drilled from each hand sample. Remanent magnetization of most specimens were measured using a cryogenic magnetometer (ScT C-112). A spinner magnetometer (Schonstedt SSM-1A) was used for measurement of some tuffaceous samples which had relatively high magnetic intensity.

Magnetic cleaning was carried out by means of alternating field (AF) demagnetization and thermal demagnetization. One pilot specimen from each site was subjected to progressive AF demagnetization generally up to 100 mT. Stepwise thermal demagnetization up to about 500° C was also carried out for at least one specimen from each site. The residual field inside the thermal demagnetization apparatus was controlled to be less than 10 nT. Results of the progressive demagnetization tests were analyzed using vector end point orthogonal projections. Typical examples of the demagnetization behaviors are shown in Fig. 3.

The samples from the Morozaki Group had the magnetic intensity typically of the order of 10^{-3} A/m before demagnetization. The remanent magnetization showed significant changes in direction after beginning steps of the progressive demagnetization, that is, at 10 mT of AF or at the temperature about 150° C. The trajectory of the magnetization vector of further demagnetization steps was linear toward the origin (Fig. 3[a], [b]). While thermal treatments above temperature of 550° C generally allowed complete demagnetization, the remanent magnetization in some samples were distributed in a high coercivity range over 100 mT. The results of the progressive AF and thermal demagnetization tests suggest that the remanent magnetization essentially consists of a single component with secondary overprint of soft magnetization. The directions of the magnetic component observed by AF and thermal treatments are similar within sites. The magnetization was, therefore, assumed the characteristic component of the Morozaki Group.

The magnetic intensities of the samples from the Mizunami Group widely range between the order of 10^{-2} to 10^{-5} A/m. The higher intensity values are of tuffaceous sediments. The demagnetization behavior of some samples resembles to that of the Morozaki Group, suggesting residence of an essentially single component (Fig. 3[d]). The vector trajectories of most samples, however, are a little more complex. Significant amount of secondary magnetization seems to remain until the demagnetization at 40 mT AF or over the temperature about 300° C (Fig. 3[c]). The linear decay of the magnetic vector toward the origin is observed in the higher demagnetization

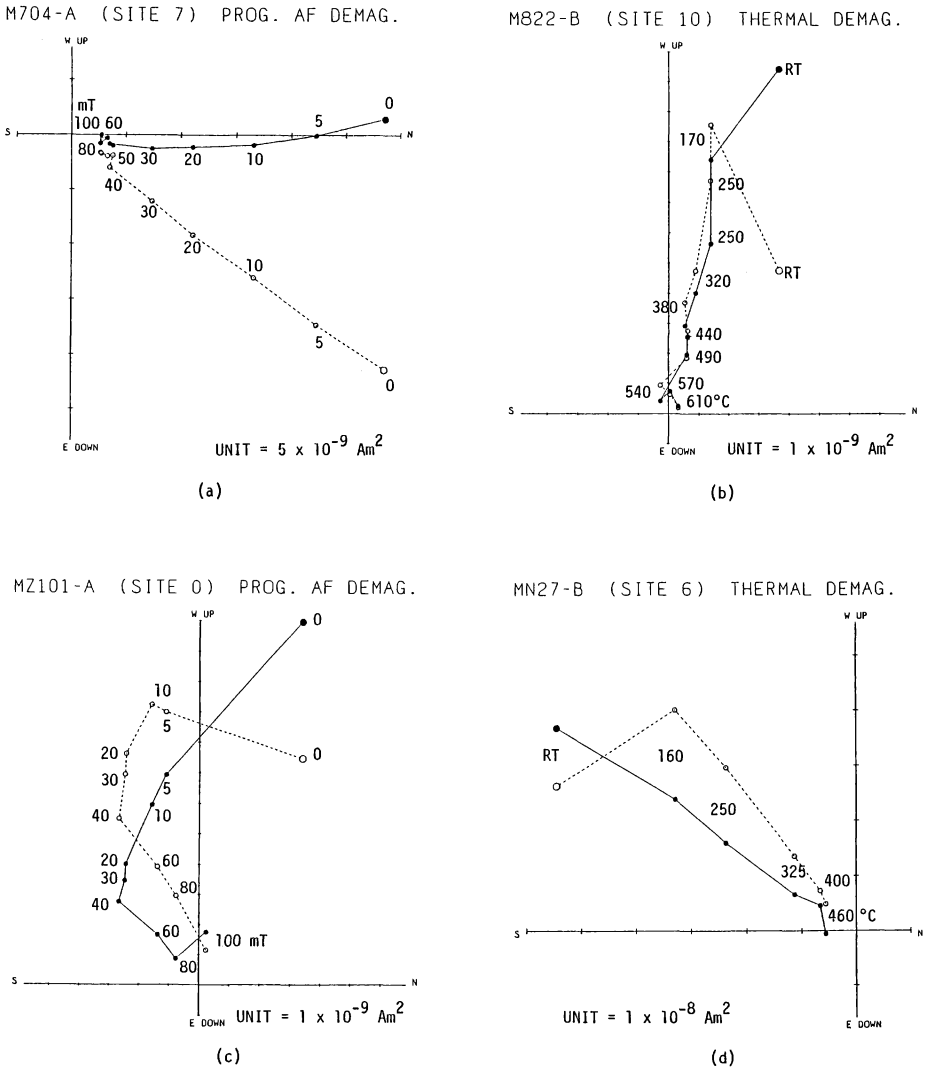


Fig. 3. Orthogonal projection diagrams of progressive demagnetization. Solid (open) symbols indicate projection of magnetic vector end point on horizontal (vertical) planes, before tilt correction. (a) and (b): Morozaki Group; (c) and (d): Mizunami Group.

steps. Since the low coercivity or low temperature component generally directed to the north and downward, its origin can be assumed the viscous remanent magnetization acquired under the recent geomagnetic field. Mudstone samples from two sites of the Oidawara Formation had very weak magnetic intensities of the order of 10^{-5} A/m. The remanent magnetization showed erratic changes during the progressive demagnetization, especially in the temperature range over 300° C. Since the higher temperature component was not detected, these two sites were abandoned in the following procedures.

Routine magnetic cleaning of additional samples were made using AF demagnetization. The optimum demagnetizing fields, which allow to isolate the characteristic magnetic directions, were determined as 40 mT for the Morozaki Group, and as 60 mT for the Mizunami Group. The essentially same directions were also detected by the thermal demagnetization of the pilot specimens at the temperatures above 250°C or 300°C. Therefore, individual sample magnetizations, determined by either the AF or the thermal method, were combined to obtain site mean directions of the characteristic magnetization.

The mean magnetic directions and the precision parameters of 14 sites of the Morozaki Group and 8 sites of the Mizunami Group are listed in Table 1, and also shown in Fig. 4. One site of the Mizunami Group, which had highly divergent magnetic directions, was rejected. The magnetic directions listed in Table 1 have been

Table 1. Site mean paleomagnetic data after tilt correction.

Site	Remanence		Statistics			Polarity	VGP	
	D	*I*	α_{95}	*N*	*k*		Lat.	Long.
Morozaki Group								
6	29.8	66.1	4.2	7	208.8	N	64.1	-174.2
3	56.8	48.6	10.7	7	33.1	N	42.3	-143.6
5	-12.1	51.9	16.4	7	14.5	N	79.7	38.1
7	24.6	29.8	23.4	5	11.6	N	61.1	-99.0
4	34.4	48.3	9.0	7	46.2	N	60.5	-132.3
2	16.3	55.5	14.2	7	19.1	N	76.7	-143.5
1	49.9	59.7	4.6	7	171.3	N	50.6	-156.9
8B	-12.9	8.9	38.6	4	6.6	I	—	—
8A	-2.9	6.3	23.5	6	9.1	I	—	—
9	-116.7	-36.7	10.8	8	27.1	R	33.0	-136.3
13	-145.0	-62.8	9.2	7	44.2	R	61.6	-163.3
10	-103.7	-58.7	8.9	5	74.8	R	30.8	-162.2
12	-132.6	-60.1	9.2	3	182.3	R	52.5	-157.2
11	-118.4	-59.6	13.4	6	26.0	R	41.8	-159.1
Mizunami Group								
6	-150.4	-42.3	10.9	8	26.7	R	62.3	-117.7
11	-162.8	-70.7	27.5	4	12.1	R	67.1	163.0
2	-117.8	-46.1	23.0	6	9.4	R	37.2	-142.6
1	-108.5	-18.3	36.3	5	5.4	I	—	—
0U	-127.7	-47.6	13.7	5	32.0	R	45.7	-139.4
0D	-115.8	-44.3	21.8	5	13.2	R	35.0	-142.0
12	-119.3	-44.5	28.8	8	4.6	R	38.0	-140.8
13	38.2	35.3	22.7	5	12.4	N	52.9	-117.7

D and *I*: site mean declination and inclination in degrees.
α_{95}: radius of the 95% confidence circle in degrees.
N: number of the specimens.
k: precision parameter of FISHER (1953).
Polarity: N (Normal), I (Intermediate) and R (Reversed).

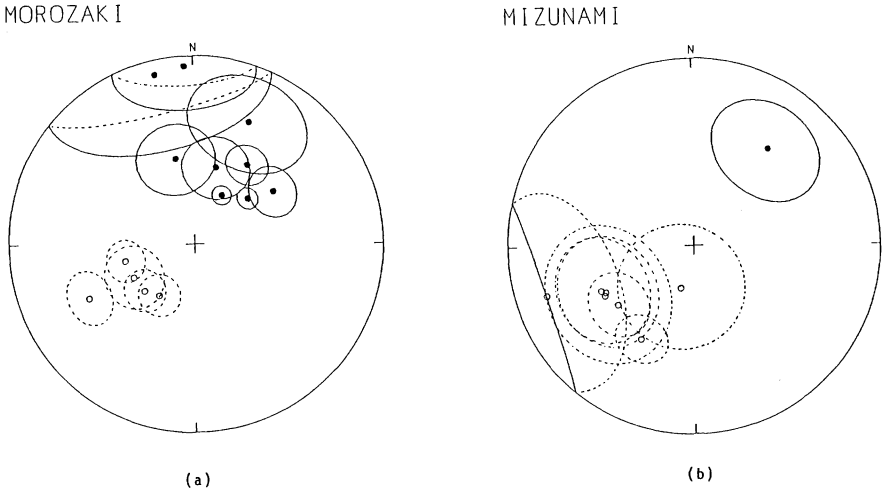


Fig. 4. Site mean directions and their 95% confidence circles of characteristic magnetization of the Morozaki Group (a) and the Mizunami Group (b), after tilt correction. Solid (open) symbols are on the lower (upper) hemisphere of equal-area projection.

obtained after correction for the tilt of strata (typically less than 15°). The magnetic directions can be assumed, therefore, to represent directions of the ancient geomagnetic field at the time of the deposition. While several sites showed intermediate directions with low inclination values and relatively large radius of 95% confidence circles (α_{95}), most of the site mean directions can be classified into normal and reversed polarity. Clockwise deflection of the declinations is observed in both of the Morozaki Group and the Mizunami Group.

5. Discussion

5.1 Magnetostratigraphy

The stratigraphic distribution of normal and reversed polarity sites is shown in Fig. 2. The Morozaki Group is divided into at least two polarity zones, while the two sites in the middle part are magnetized in rather intermediate directions. The Mizunami Group, on one hand, has remanent magnetization of reversed polarity with an exception of a single site in the lower part. The magnetic polarity sequences shown in Fig. 2 generally support the biostratigraphic correlation between the Morozaki Group and the Mizunami Group, and to the Ichishi Group. A possible age assignment of these groups is shown in Fig. 5.

The reversed polarity sites are dominant in the Toyohama Formation and the lowermost part of the Utsumi Formation of the Morozaki Group, and in the Akeyo Formation of the Mizunami Group. The micropaleontological data (IBARAKI *et al.*, 1984; IBARAKI, 1981) show that these formations are correlative to the Blow's Zones N6 to N8 of the late Early Miocene. The Ichishi Group is also suggested to be included in the same period; HAYASHIDA and ITO (1984) assigned the middle to the upper parts of the Ichishi Group to Epoch 16, a reversed polarity interval dated from 17.6 to 15.2

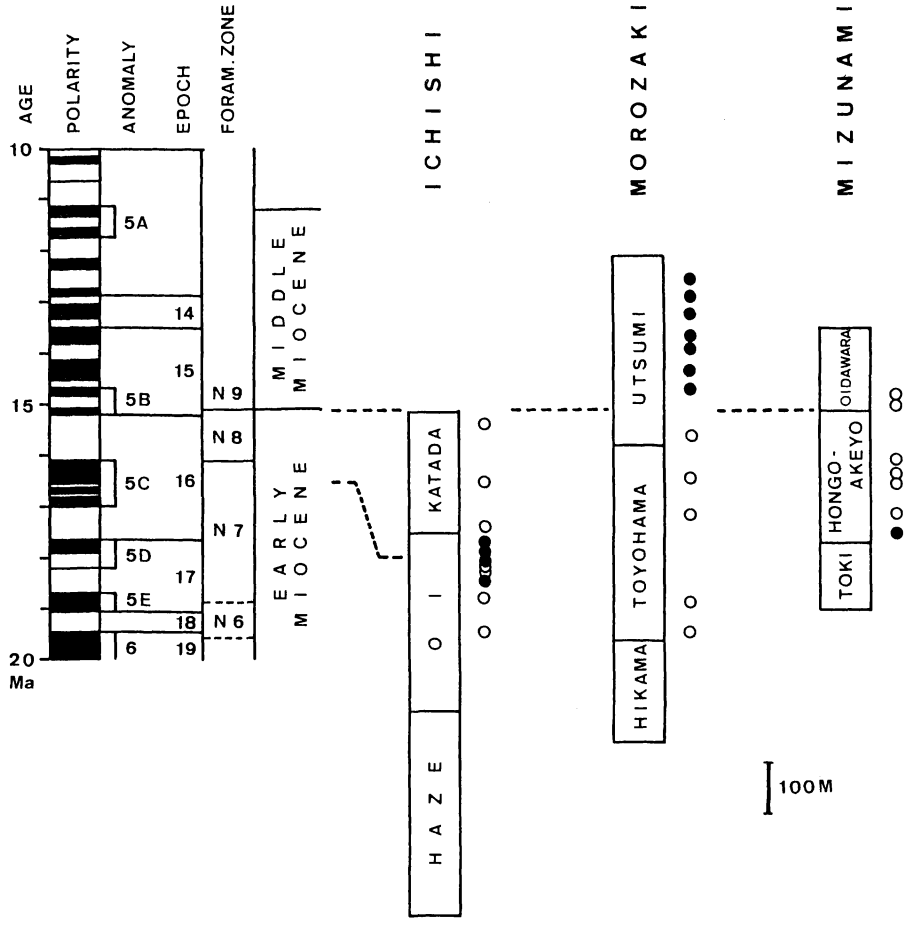


Fig. 5. A possible correlation of the Ichishi Group, the Morozaki Group and the Mizunami Group. Solid and open circles show normal and reversed polarity sites (HAYASHIDA and ITO, 1984; this study). Paleomagnetic polarity timescale and biostratigraphic datum levels are after HSÜ *et al.* (1984).

Ma (LOWRIE and ALVAREZ, 1981), based on its correlation to the Zone N7 and N8. It is suggested, therefore, that the middle parts of the Morozaki and Mizunami Groups were deposited during Epoch 16.

A normal polarity zone in the upper part of the Utsumi Formation of the Morozaki Group is correlative to the normal polarity interval following Epoch 16. The Utsumi Formation yielded *Orbulina universa* in the uppermost part, and was correlated to the Zone N9 (DOI, 1984). OPDYKE *et al.* (1974) showed that the first appearance of *Orbulina*, which defines the N9/N8 boundary and corresponds to the Early/Middle Miocene boundary, occurs around the horizon of a magnetic polarity change from Epoch 16 to Epoch 15. Recent paleomagnetic studies in the Deep Sea Drilling Project, utilizing hydraulic piston corers, confirmed the horizon of the *Orbulina* datum (HSÜ *et al.*, 1984); the datum was found near the bottom of a normal

polarity zone, which was identified as the chron C-5B of the standard magnetic polarity timescale. Although the exact horizons of the *Orbulina* datum and the magnetic polarity changes are not determined in the Utsumi Formation yet, it is acceptable that the upper part of the Utsumi Formation was deposited during some normal polarity interval in Epoch 15.

The polarity data from the Oidawara Formation are not enough to clarify its magnetostratigraphic correlation. The two sites situated on the lowermost part of the Oidawara Formation had reversed magnetic polarity, and the upper two sites were rejected from the determination of polarity because of weak magnetic intensity. The Oidawara Formation was shown to be included in the *Denticulopsis lauta* Zone of diatom biostratigraphy (KOIZUMI, 1981), which correspond to the Zone N9. It should be assumed, therefore, that the Oidawara Formation was formed during Epoch 15, correlating the reversely magnetized horizon to some reversed polarity episode.

5.2 *Rotation of Southwest Japan*

The 7 normal and 5 reversed polarity sites of the Morozaki Group yield the over-all mean direction: *Dm* (mean declination)=40.3°, *Im* (mean inclination)=55.5° with α_{95}=9.7°. The mean direction of 2 normal and 5 reversed polarity sites of the Mizunami Group is given as follows: *Dm*=48.4°, *Im*=48.3° with α_{95}=11.8°. Both mean directions show significant clockwise deflection in declinations from the present north direction, while the mean inclinations are comparable to those expected from the present latitudes of the sampling areas. Since the late Tertiary paleomagnetic poles for Eurasia are almost coincident with the present north pole (IRVING and IRVING, 1984), the deflected magnetic directions of the Morozaki and Mizunami Groups can be attributed to local tectonic rotation relative to Eurasia. The appropriate tilt correction of the magnetic directions implies that the rotation occurred around a vertical axis.

When the site mean magnetic directions are compared between the lower and upper part of the sections, a possibility that the rotation may have occurred during the deposition of the Morozaki and the Mizunami Group is suggested. The magnetostratigraphic argument draws a conclusion that the Morozaki Group is divided into the reversed and normal polarity zones, which are assigned to the late Early Miocene and to the early Middle Miocene, respectively. Based on the biostratigraphic data, the Mizunami Group is also subdivided into the same stages, that is, the Hongo and Akeyo formations in the lower part and the disconformably overlying Oidawara Formation. Formation mean magnetic directions were obtained respectively for the lower and upper members of the Morozaki and the Mizunami Group, as listed in Table 2. The deflection of the mean declinations in the lower members exceeds 50°, and that of the upper members is less than 30°. The difference of the rotation angles between the lower and upper means is not significant in 95% confidence level. Especially, the mean magnetic direction of the Oidawara Formation, which was calculated from merely two site means, has a great ambiguity. It is remarkable, however, that the magnetic directions of each stage are concordant between the Morozaki Group and the Mizunami Group.

Paleomagnetic directions concordant with the present results are observed in the Setouchi Miocene Series in the other areas. The Ichishi Group, correlative to the Zone N7 and N8, shows clockwise declination shift about 47° from the present north

Table 2. Formation mean paleomagnetic data.

Formation (age)	Location		N	Direction			VGP Position			Ref.
	Lat.	Long.		*Dm*	Im	α_{95}	Lat.	Long.	A_{95}	
Early Middle Miocene										
Morozaki (N9)	34.7	136.9	7	28.4	53.3	13.6	66.6	-142.2	16.4	1
Mizunami (N9)	35.4	137.2	2	25.8	56.6	—	70.0	-153.1	—	1
Takakura (N9?)	35.1	134.0	4	23.5	37.6	11.4	65.0	-107.9	8.8	2
Mean			3				68.4	-133.1	14.4	
Late Early Miocene										
Morozaki (N6–7)	34.7	136.9	5	58.0	56.4	12.9	44.4	-154.7	14.8	1
Mizunami (N8)	35.4	137.2	5	55.1	44.1	8.9	42.1	-137.5	9.9	1
Ichishi (N7–8)	34.7	136.4	11	45.1	48.8	11.9	51.9	-142.3	14.3	3
Hokusetsu (?)	35.1	137.6	4	48.0	50.0	14.0	49.8	-140.2	17.2	2
Tonosho (?)	34.5	134.2	4	42.2	60.3	11.1	56.3	-160.3	15.8	2
Mean			5				49.2	-146.5	8.1	
Oligocene-Miocene										
Hata (21–23 Ma)	35.1	132.5	6	69.9	49.5	14.5	32.9	-154.3	14.9	4
Kawauchi (28 Ma)	35.0	132.4	6	52.2	33.5	15.3	41.8	-130.4	14.9	4

Dm and *Im*: formation mean declination and inclination in degrees.
N: number of the sites.
α_{95} and A_{95}: radius of the 95% confidence circle in degrees.
References are: 1, this study; 2, TORII (1985); 3, HAYASHIDA and ITO (1984); 4, OTOFUJI and MATSUDA (1984).

(HAYASHIDA and ITO, 1984). TORII (1985) also observed clockwise deflected magnetic directions in some sequences of the Setouchi Miocene Series; these are Dm=48.0° of the Hokusetsu Subgroup in the Shidara area, Dm=42.2° of the Tonosho Group in the Shodo-shima and Te-shima Islands, and Dm=23.5° of the Takakura Formation in the Tsuyama basin. While micropaleontological age assignment of these three sequences are not established yet, the Takakura Formation was suggested to be younger than the others; the Setouchi Miocene Series in the Tsuyama basin was correlated to the uppermost formations of the Morozaki and Mizunami Groups, based on an idea of sedimentary cycles (ITOIGAWA and SHIBATA, 1973).

Figure 6 shows plots of the virtual geomagnetic pole (VGP) positions determined from the Setouchi Miocene Series and the volcanic rocks in the San'in district (See Table 2). The paleomagnetic poles of the younger stage of the Setouchi Miocene Series, which is correlative to the Zone N9 of the early Middle Miocene, locate close to the present north pole in comparison with the Early Miocene poles (Fig. 6[a]). This fact implies that the rotation of Southwest Japan had partly occurred until the stage of the Zone N9 of the early Middle Miocene. The paleomagnetic poles of 21–23 Ma and 28 Ma (OTOFUJI and MATSUDA, 1984) are plotted on the region almost the same

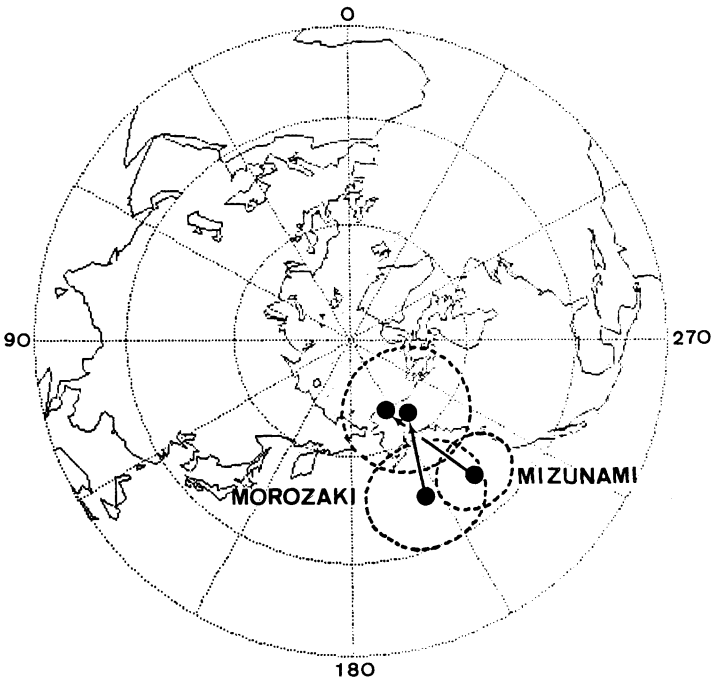
as the older poles of the Setouchi Miocene Series (Fig. 6[b]). This fact suggests that no significant rotation of Southwest Japan had occurred until the late Early Miocene. Thus, the paleomagnetic directions of the Setouchi Miocene Series may define the beginning of the rotation of Southwest Japan.

5.3 Opening of the Japan Sea

The clockwise rotation of Southwest Japan has been attributed to fan-shape opening of the southwestern part of the Japan Sea (OTOFUJI and MATSUDA, 1983; HAYASHIDA and ITO, 1984). The clockwise deflection of the paleomagnetic directions could be explained by other tectonic models, such as rotation of small structural domains along a transform fault zone (e.g., BECK, 1976; WELLS and COE, 1985). However, the paleomagnetic directions of the Setouchi Miocene Series distributed in several sedimentary basins suggest coherent rotation of Southwest Japan. Geologic structure of the pre-Neogene basement of Southwest Japan also seems to eliminate the possibility of the rotation of separate blocks in the Miocene (OTOFUJI *et al.*, 1985). Thus, the drift of Southwest Japan with the dimension of a micro-plate size is more preferred as the cause of the deflection of the paleomagnetic directions. The comparison of the paleomagnetic directions between Southwest Japan and the eastern margin of the Asian Continent is desirable; the Miocene poles for the Korean Peninsula or Northeastern China are not determined yet, however. The Tertiary poles for the North and South China blocks are still in preliminary state (LIN *et al.*, 1985). Assuming that the Neogene motion of the eastern margin of the Asian continent relative to the pole was negligibly small, the rotation of Southwest Japan can be attributed to the opening of the Japan Sea.

The paleomagnetic directions of the Setouchi Miocene Series suggested the possibility that the rotation of Southwest Japan began at around the late Early Miocene/early Middle Miocene boundary (N8/N9 boundary). This possibility implies that Southwest Japan had been situated adjacent to the Asian continental margin until the late Early Miocene, and proceeded the drift in the early Middle Miocene. The numerical age of the Early Miocene/Middle Miocene boundary (*Orbulina* datum) is estimated at 15.1 Ma (HSÜ *et al.*, 1984). The present result thus confirms the timing of the opening of the Japan Sea, 14.7 Ma, proposed by OTOFUJI *et al.* (1985).

The timing of the opening inferred from the paleomagnetic data of the Setouchi Miocene Series is well correlated with the geological records represented by the Neogene sequences around the Japan Sea. The andesitic volcanic rocks ("Green Tuffs") and non-marine deposits (e.g., the Daijima Formation) of the late Oligocene to the Early Miocene age have been regarded as the products of the initial rifting in the Japan Sea (UYEDA and MIYASHIRO, 1974). The overlying marine sediments of the late Early Miocene (Nishikurosawa Stage; Zone N8) are assumed to evidence the commencement of the marine transgression in the Japan Sea (INGLE, 1975). The paleomagnetic data suggest that until this stage the drift of Southwest Japan had not proceeded significantly yet. The sediments of the subsequent periods (Zone N9 or the younger) were suggested to show the increasing subsidence of the basin (INGLE, 1975). This stage corresponds to the period when the rotation of Southwest Japan was advanced.

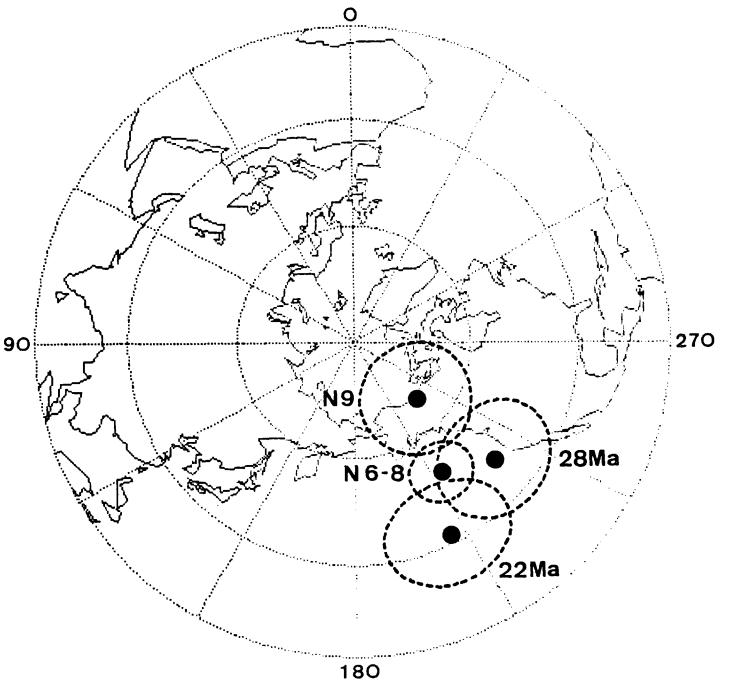


(a)

Fig. 6. Virtual geomagnetic pole (VGP) positions plotted on the Northern Hemisphere of equal-area projection (See Table 2). (a) VGP's of the Morozaki and the Mizunami Group. Transition of the poles from the late Early Miocene to the early Middle Miocene are shown by arrows. (b) VGP positions determined from the volcanic rocks in the San'in district (Kawauchi: 28 Ma, Hata: 22 Ma), and the Setouchi Miocene Series (late Early Miocene: N.6–8, early Middle Miocene: N.9). Confidence circles of the 95% level are shown except for the younger pole of the Mizunami Group (a).

6. Conclusion

Paleomagnetic studies of the Morozaki Group and the Mizunami Group in the eastern part of Southwest Japan showed that occurrence of a clockwise tectonic rotation after the late Early Miocene. The concordant magnetic directions are observed in several basins of the Setouchi Miocene Series, and the deflection can be attributed to the coherent rotation of Southwest Japan. While the lower members of the Setouchi Miocene Series and the volcanic rocks in the San'in district older than 20 Ma show the declination shift about 50°, the deflection in the upper members is less than 30°. This fact suggests that Southwest Japan began to rotate during the deposition of the Setouchi Miocene Series. Nearly a half of the rotation might be attained in the stage of Zone N9 of the early Middle Miocene. The timing of the rotation of Southwest Japan is coincident with the increasing subsidence in the Japan Sea, which is implied from the stratigraphic records on its coastal areas.



(b)

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